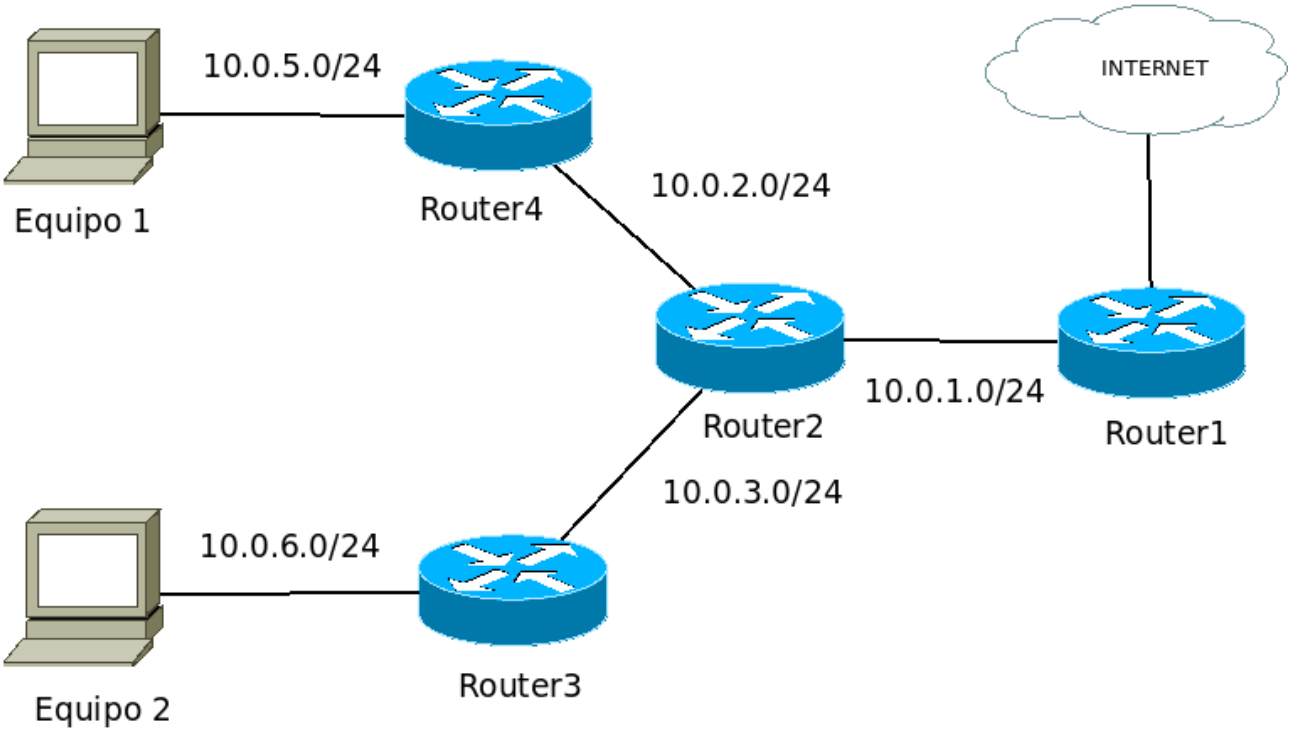
UT1. PRÁCTICA DE ENCAMINAMIENTO

1. Tenemos la siguiente red.



Genera el direccionamiento y las tablas de encaminamiento para comunicar todos los equipos y todos los routers.

Direccionamiento

Equipo	Interfaz	IP/red	Gateway	Servidor Nombres
Router1	**enp0s9 (internet)**	172.31.7.1/16	172.31.0.1/16	8.8.8.8
	enp0s10 (interna)	10.0.1.1/24		
Router2	**enp0s8(router1)**	10.0.1.2/24	10.0.1.1/24	
	enp0s9 (router4)	10.0.2.1/24	10.0.2.2/24	
	enp0s10(router3)	10.0.3.1/24	10.0.3.2/24	
Router3	**enp0s9 (router4)**	10.0.3.2/24	10.0.3.1/24	
	enp0s10(equipo2)	10.0.6.1/24		
Router4	**enp0s9 (router4)**	10.0.2.2/24	10.0.2.1/24	
	enp0s10(equipo1)	10.0.5.1/24		
Equipo1	**enp0s9 (router4)**	10.0.5.2/24	10.0.5.1/24	
Equipo2	**enp0s9 (router3)**	10.0.6.2/24	10.0.6.1/24	

Tabla de encaminamiento routers

Router	Destino	Gateway	Interfaz
Router1	10.0.1.0/24	10.0.1.2/24	enp0s10
	10.0.2.0/24	10.0.1.2/24	enp0s10
	10.0.3.0/24	10.0.1.2/24	enp0s10
	10.0.5.0/24	10.0.1.2/24	enp0s10
	10.0.6.0/24	10.0.1.2/24	enp0s10
Router2	10.0.1.0/24		enp0s8
	10.0.2.0/24		enp0s9
	10.0.3.0/24	10.0.2.2	enp0s10
	10.0.5.0/24	10.0.3.2	enp0s9
	10.0.6.0/24	10.0.1.1	enp0s10

2. Simula el esquema de la red con equipos linux (sin interfaz gráfica). Genera el direccionamiento y el NAT para que todos los equipos tengan conexión entre sí y salgan a internet.

Solución:

Nota. Todas las redes pueden estar dentro de la misma red interna, menos la que sale a internet que es adaptador puente.

Router 1.

usuario@Router1:~$ cat /etc/netplan/00-installer-config.yaml

```
network:
   ethernets:
      enp0s3:
         dhcp4: true
      enp0s8:
         dhcp4: true
      enp0s9:
         dhcp4: false
         addresses:
          - 10.0.1.1/24
         nameservers:
          addresses:
            - 1.1.1.1
            - 8.8.8.8
         routes:
          - to: 10.0.2.0/24
            via: 10.0.1.2
          - to: 10.0.3.0/24
            via: 10.0.1.2
          - to: 10.0.5.0/24
            via: 10.0.1.2
          - to: 10.0.6.0/24
            via: 10.0.1.2
      enp0s10:
         dhcp4: true
   version: 2
```

Router 2

```
network:
   ethernets:
      enp0s3:
         dhcp4: true
      enp0s8:
         dhcp4: false
         addresses:
          - 10.0.1.2/24
         nameservers:
          addresses: [1.1.1.1, 8.8.8.8]
         routes:
          - to: default
            via: 10.0.1.1
      enp0s9:
         dhcp4: false
         addresses:
          - 10.0.2.1/24
         nameservers:
          addresses: [1.1.1.1, 8.8.8.8]
         routes:
          - to: 10.0.5.0/24
            via: 10.0.2.2
      enp0s10:
         dhcp4: false
         addresses:
          - 10.0.3.1/24
         nameservers:
          addresses: [1.1.1.1, 8.8.8.8]
         routes:
          - to: 10.0.6.0/24
            via: 10.0.3.2
   version: 2
```

Router3

```
network:
   ethernets:
      enp0s3:
         dhcp4: true
      enp0s8:
         dhcp4: false
         addresses:
          - 10.0.3.2/24
         nameservers:
          addresses:
            - 1.1.1.1
            - 8.8.8.8
         routes:
          - to: default
            via: 10.0.3.1
      enp0s9:
         dhcp4: false
         addresses:
          - 10.0.6.1/24
      enp0s10:
         dhcp4: true
   version: 2
```

Equipo2

```
network:
   ethernets:
      enp0s3:
         dhcp4: true
      enp0s8:
         dhcp4: false
         addresses:
          - 10.0.6.2/24
         nameservers:
          addresses:
            - 1.1.1.1
            - 8.8.8.8
         routes:
          - to: default
            via: 10.0.6.1
      enp0s9:
         dhcp4: true
      enp0s10:
         dhcp4: true
   version: 2
```

Conseguir NAT.

Después de configurar todo para que todos los routers se vean entre si y se comuniquen todos los equipos lo que hay que conseguir es que todas las redes salgan a internet a través de NAT.

```
sudo iptables -t nat -L -nv
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

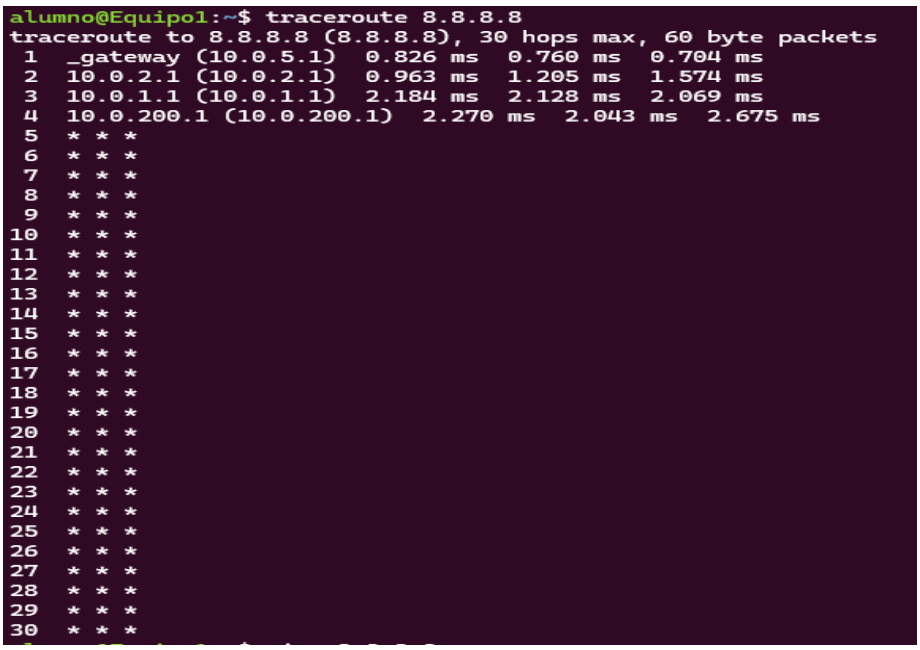
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 24 packets, 3100 bytes)
 pkts bytes target     prot opt in     out     source               destination
    8   604 MASQUERADE  0    --  *      enp0s8  10.0.1.0/24       0.0.0.0/0
    7   528 MASQUERADE  0    --  *      enp0s8  10.0.2.0/24       0.0.0.0/0
    8   584 MASQUERADE  0    --  *      enp0s8  10.0.3.0/24       0.0.0.0/0
    7   528 MASQUERADE  0    --  *      enp0s8  10.0.5.0/24       0.0.0.0/0
    8   584 MASQUERADE  0    --  *      enp0s8  10.0.6.0/24       0.0.0.0/0


sudo cat  /etc/iptables/rules.v4
# Generated by iptables-save v1.8.10 (nf_tables) on Tue Sep 29 10:01:42
2026
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
:POSTROUTING ACCEPT [16:1028]
-A POSTROUTING -s 10.0.1.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.2.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.3.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.5.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.6.0/24 -o enp0s8 -j MASQUERADE
COMMIT
# Completed on Tue Sep 29 10:01:42 2026
```

Traceroute

Equipo1:



Equipo2:

